5G Radio Planning

Previous preparation

- Read description of Radio Mobile application.
- Read this guide carefully.

Objectives

You are tasked with designing a **private 5G mobile radio network for a geographic area of your choice** using the Radio Mobile application.

The network will operate in the **3850-3950 MHz frequency band** and will initially consist of **three Base Stations (sites)**, each with an output power of 4 W. You should first deploy these three Base Stations and analyze the resulting radio coverage. Based on this analysis, you must then **identify suitable locations for two additional sites** in order to improve the overall coverage of the selected area.

The **Radio Mobile** application is a tool used to predict the performance of a radio system. It uses **digital terrain elevation data** to automatically extract the path profile between a transmitter and a receiver. This data is combined with environmental and statistical parameters to feed the **Irregular Terrain Model (ITS)** radio propagation model.

The ITS model of radio propagation for frequencies between **20 MHz and 20 GHz** (the Longley-Rice model) is a general-purpose model that can be applied to a wide range of engineering problems. The model, which is based on electromagnetic theory and on statistical analyses of both terrain features and radio measurements, predicts the **median attenuation of a radio signal as a function of distance** and the variability of the signal in time and in space.

Elevation data is also used to produce **altimetry maps in the background**, which can be merged with other maps from various sources, adding populated places, roads, and other geographical features to facilitate location in the area.

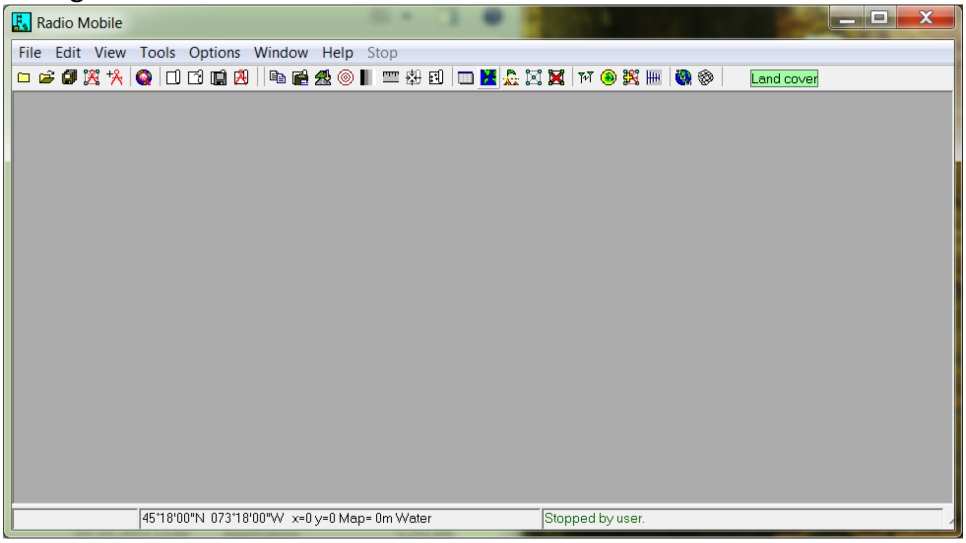
This laboratory work consists of **four major steps**:

a) Radio Mobile setup and map configuration;
b) Initial network configuration;
c) Initial coverage analysis;
d) Network coverage improvement.

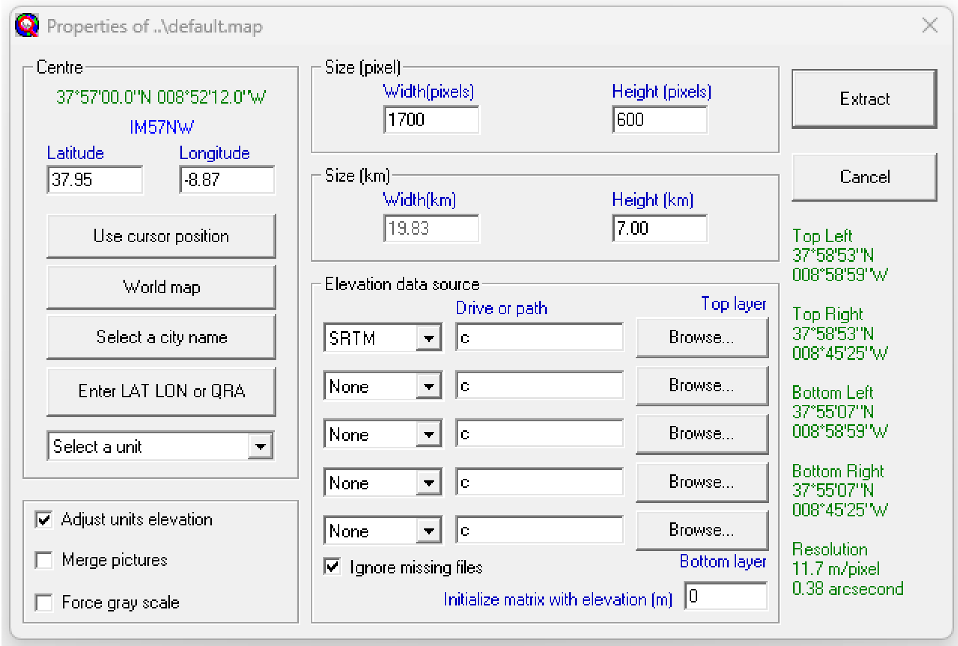
At the end of the laboratory work, you should prepare a **concise report** and submit it via **Moodle**.

Step A – Radio Mobile setup and map configuration

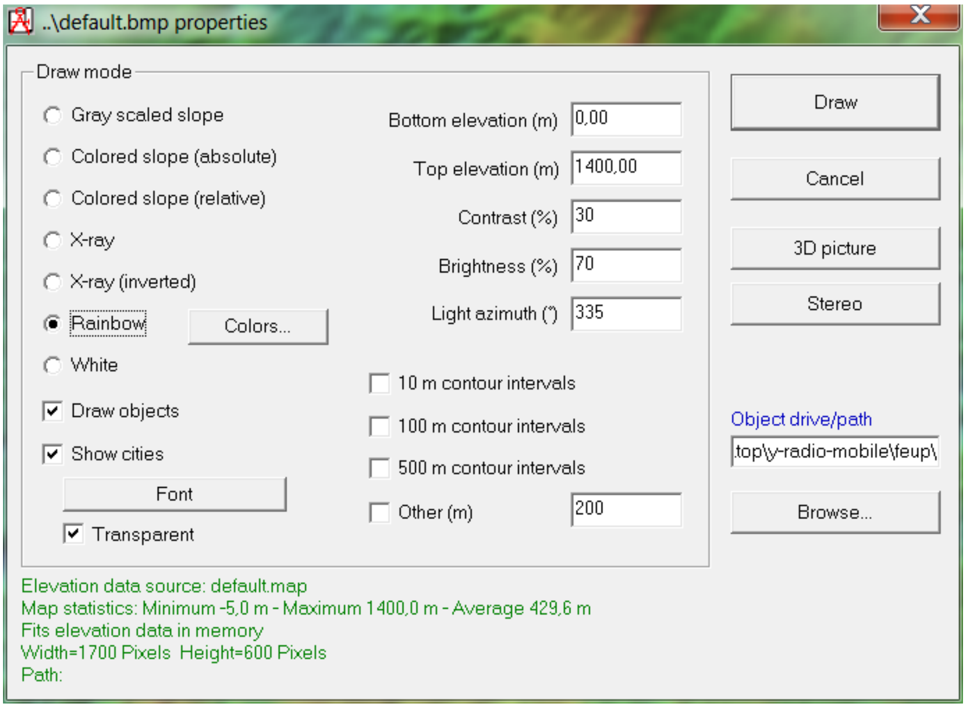
1. Install the Radio Mobile application on your PC. If you download the software from Moodle, you should extract the file *rm-software.7z*. If you download from the Radio Mobile website, you should follow the website instructions.
2. Open the application by selecting the file *rmweng.exe*. You should obtain the following window:



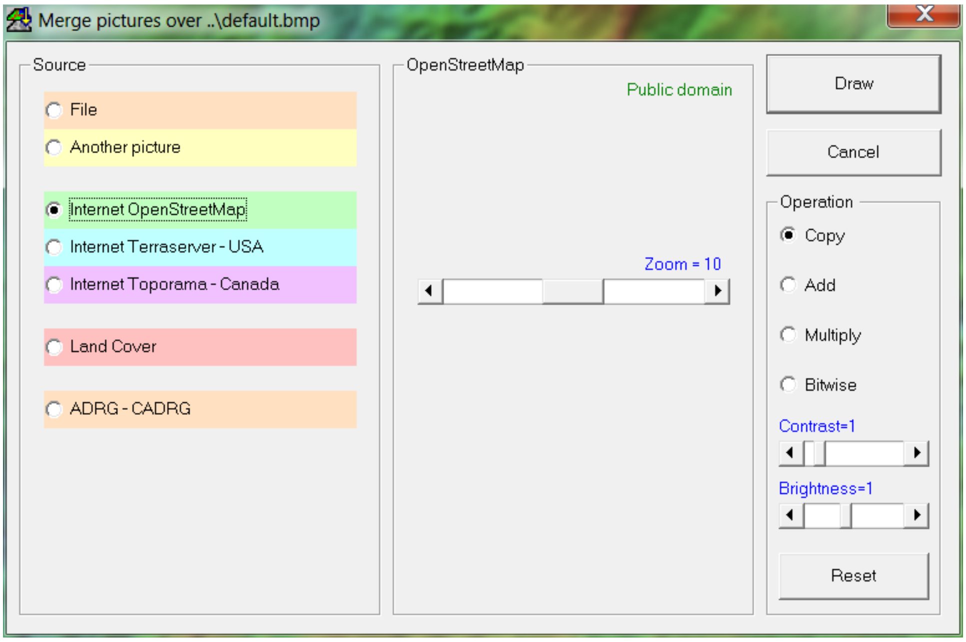
3. Define the map properties ("File" + "Map Properties"): set the centre point, size of picture (in pixels), size of map (in km), etc., according to the figure below. You will only need to **change the "Latitude" and "Longitude"** for your selected location. Finally, click on Extract.



4. Define picture properties ("File" + "Picture Properties"). This option enables the display of altimetry data.



5. Insert geographic references in the picture. “Edit” + “Merge pictures” and then “Draw”: this step enables the display of populated places and roads to facilitate the location of sites. After selecting “Draw” and then “Keep in a new picture”, you will have multiple interrelated pictures.



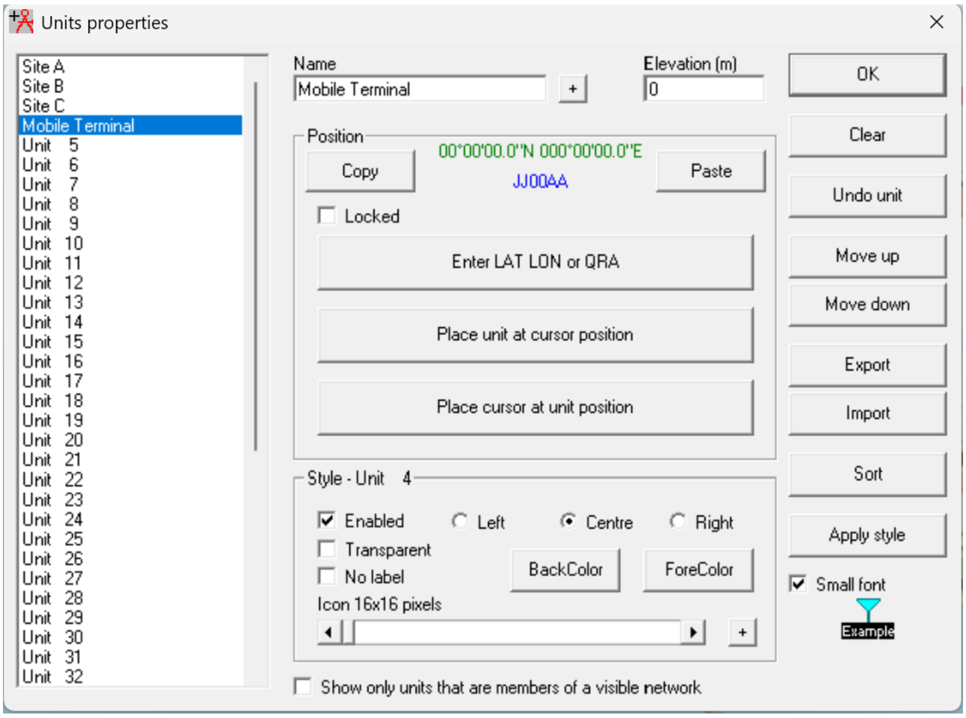
Step B - Initial network configuration

6. Enter Units properties (“File” + “Unit Properties”). The **LAT and LON** values below are provided as **examples**. You must define the locations of three initial base stations (Site A, Site B, and Site C) within the selected area.
   a) **Mobile Terminal:** input the name but no location (this will be a running unit used for coverage analysis).

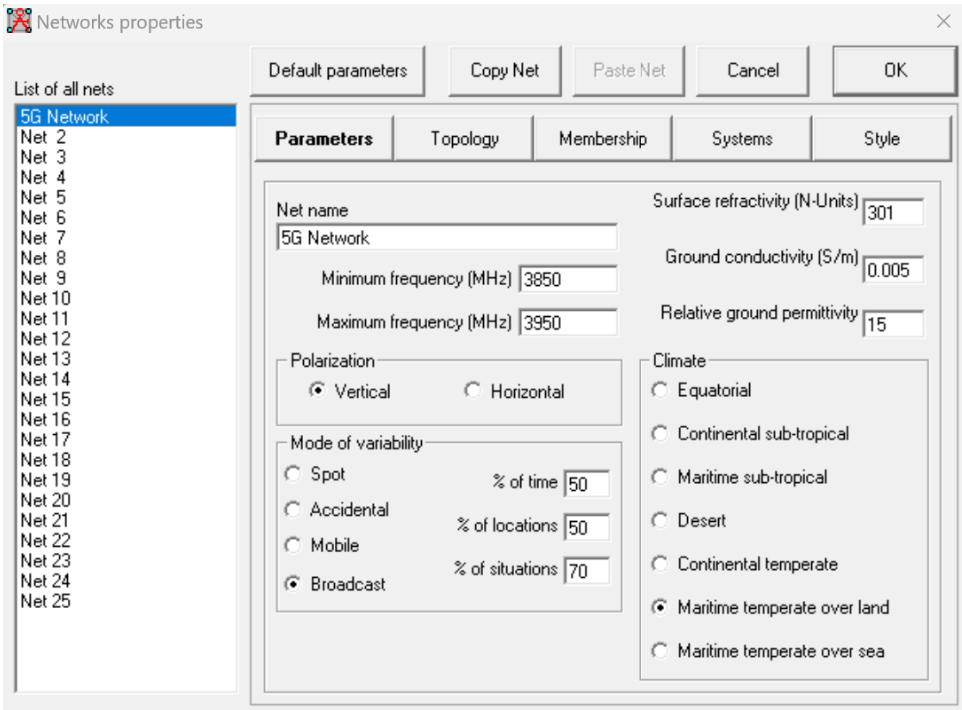
b) **Site A (change coordinates for your area)**: e.g., LAT = 37.952102; LON = -8.87669

c) **Site B (change coordinates for your area)**: e.g., LAT = 37.954082; LON = -8.879045

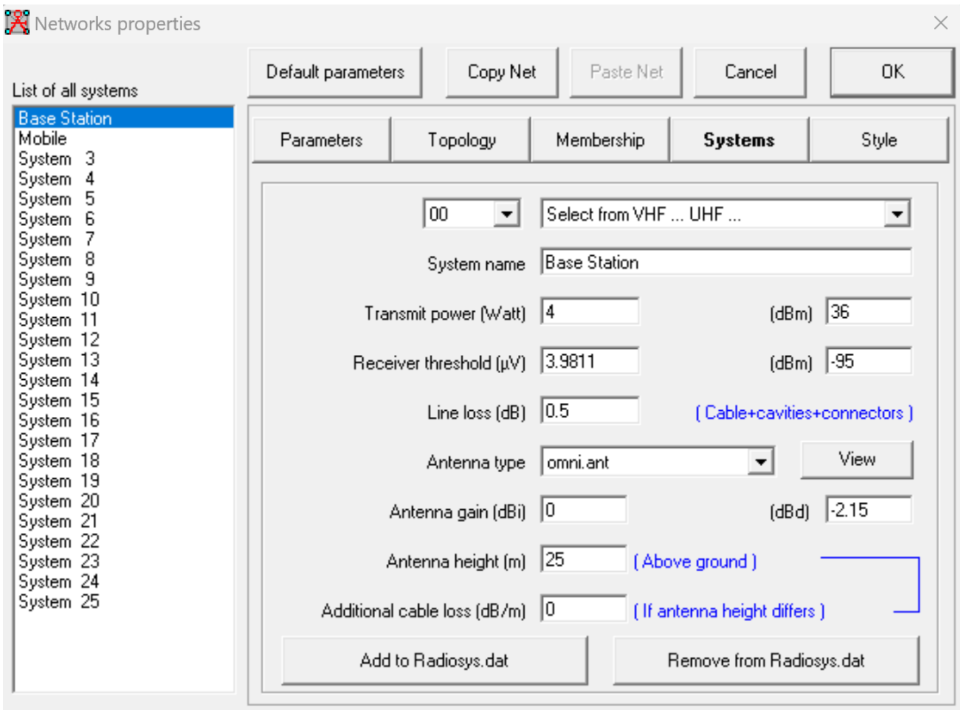
d) **Site C (change coordinates for your area)**: e.g., LAT = 37.958008; LON = -8.886982



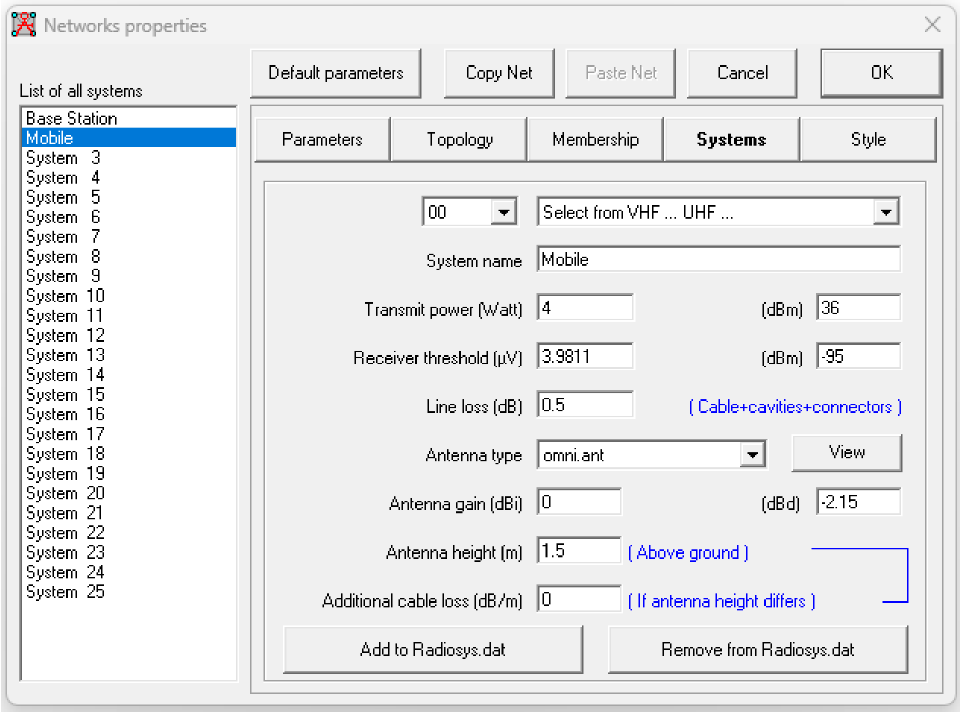
7. Define network properties ("File" + "Network Properties") → Parameters



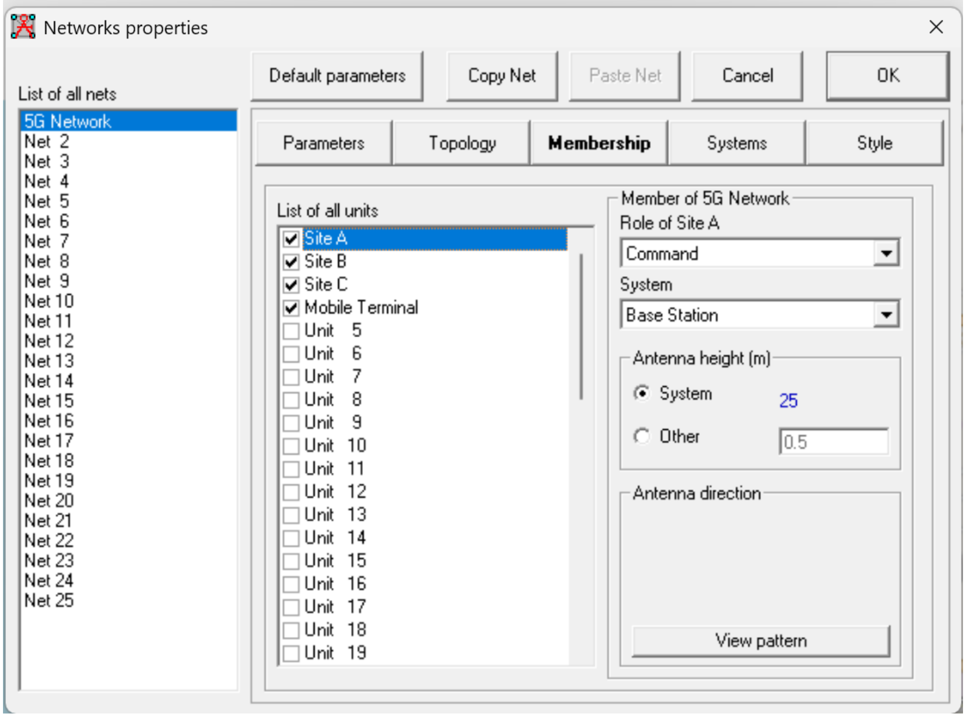
a) Base Station parameters (under "Systems")

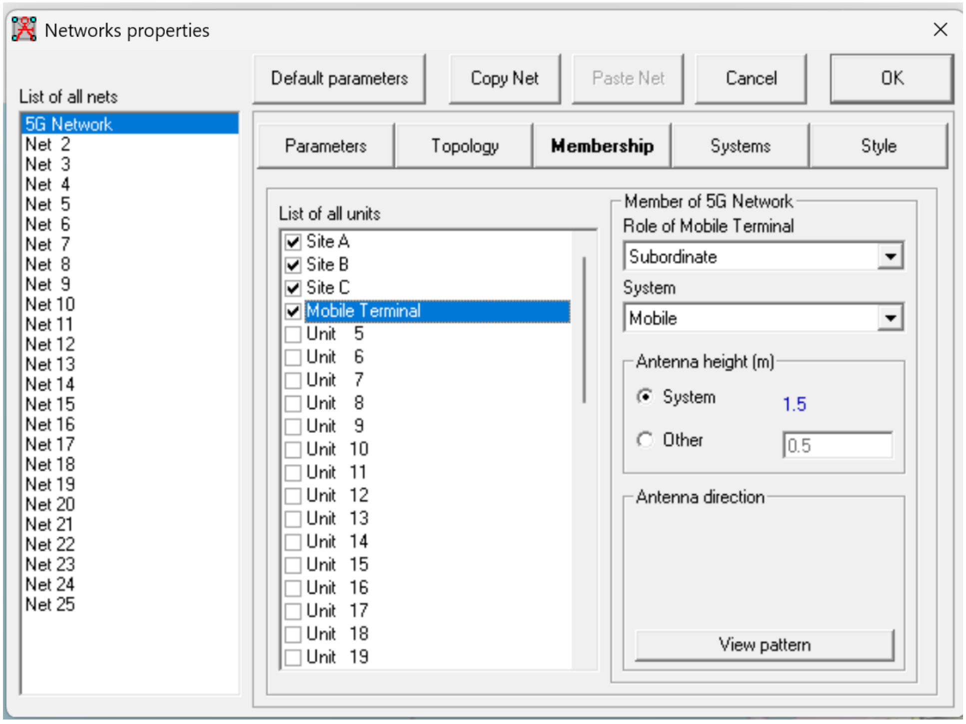


b) Mobile (5G User Equipment) parameters (under "Systems")



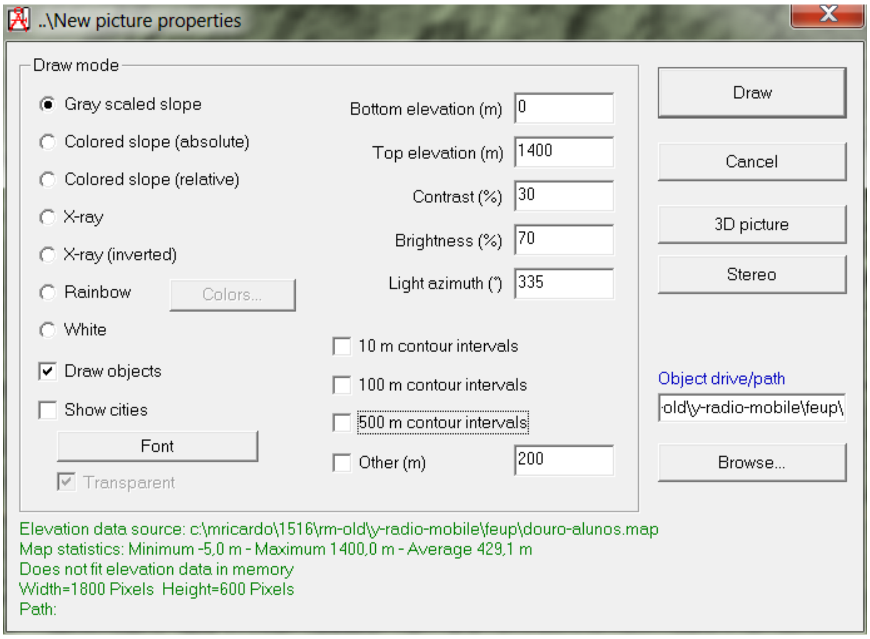
c) Membership (units vs. role+systems: **mobile terminal is subordinate** and belongs to **Mobile** system; **other units are command** and belong to **Base Station** system)



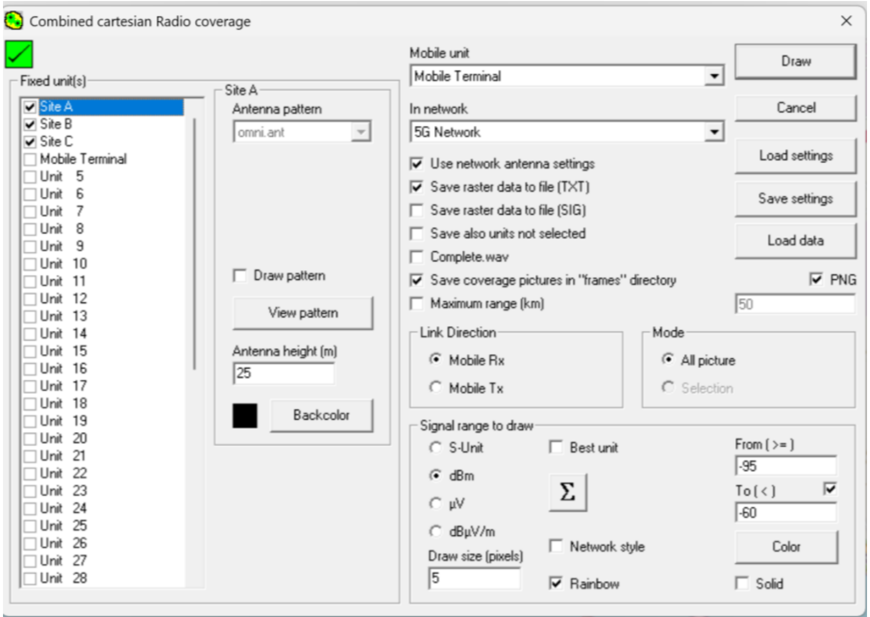


Step C – Initial coverage analysis

8. Define new picture properties ("File" + "New Picture")
   a) This new picture uses a greyscale slope to obtain suitable coverage maps
   b) This option may be used without additional geographic references, since visible slopes provide information about elevation



9. Insert geographic references in the picture, as before
    a) “Edit” + “Merge pictures” and then “Keep in actual picture” to preserve the geographic references in the subsequent coverage analysis
10. Obtain the radio coverage of the three sites
    a) “Tools” + “Radio coverage” + “Combined cartesian” and then “Keep in a new picture” + “Save coverage data”: this step allows obtaining a single map with the best received signal for each point
    b) Areas with no coverage are shown without any color, i.e., left in greyscale as in the original picture
    c) Draw size specifies the detail of the coverage, from 1 pixel to a few pixels (say 5), depending on the processing power of your computer and the stage of the analysis (initial viewing may be low resolution, say 10, but the final coverage map should have the highest possible resolution allowed by your computer)



11. Derive the cumulative distribution of radio coverage levels
    a) Consider the range value appearing in the coverage data file as $10.\log_{10}(Pr/Pr_{min})$, where Pr_{min} is -95 dBm, defined in the window above (below "From (>=)"), which is the same as the receiver sensitivity. That is, if in the coverage file appears an Rx (dB) value equal to 20 dB, that means that at the current point the power received was in fact -95 dBm + 20 dB = -75 dBm.
    b) Write a small program to read the file and obtain the cumulative coverage distribution, considering the following intervals relative to the minimum receiver sensitivity (-95 dBm): <0 <5 <10 <15 <20 <25 <30.
    c) Define the percentage of values Pr < -95 dBm (Rx < 0 dB), in the coverage file, as the percentage of area without coverage.
    d) Define the percentage of values Pr < -90 dBm (Rx < 5 dB) as the percentage of area below good coverage (5 dB margin relative to minimum receiver sensitivity)

Step D – Network coverage improvement

12. First, determine the best location for **one additional site**, resulting in a network with **four sites**. Evaluate the coverage improvement obtained with this new site.
    Then, determine the best locations for **two additional sites**, resulting in a network with **five sites**.
    The goal is to improve the overall coverage, i.e., minimize the area without coverage (or below good coverage), calculated as before.
    For this step, employ a **"What-if" mode** approach (*tentativa-erro*) to analyze the effect of candidate site locations, preferably at suitable elevations. To speed up the process, you may use lower resolution when testing alternatives.
    Obtain the cumulative distribution of radio coverage levels **when each of the two sites is added individually and when both sites are added to the network**.

Report structure

The report should not exceed **three A4 pages in 10pt font**, including text, tables, figures, and coverage maps. Coverage maps should be included right after the question they are related to. The CoverageData.txt files should not be added to the report; please upload them separately in one zipped file to the designated location in Moodle.

Questions

1) Based on the radio coverage results you obtained for the initial network with three sites (paragraphs 9 and 10), what are the percentages of:
    a) Area without coverage Pr < -95 dBm?
    b) Area below good coverage Pr < -90 dBm?
    c) Which of the coverage intervals (in dB with respect to -95 dBm: [0,5[, [5, 10[, [10,15[, [15,20[, [20,25[, [25,30[, [30, …[) defines the largest area? What is the percentage of the area that has coverage within that interval?

2) Find the best location and add one new site to the network (total of four sites). What are the coordinates of the site? What is the new percentage of area below good coverage Pr < -90 dBm? Please include the coverage map and the CoverageData.txt file.

The coordinates are : Lat = 41.19995 ; Long = -8.390871

3) Find the best location for two new sites in the network (total of five sites: note that both of the two new sites you created can be different from the new site you created in question 2). What are the coordinates of the sites? What is the new percentage of area below good coverage Pr < -90 dBm? Please include the coverage map and the CoverageData.txt file.

4) By setting the "Draw size" resolution to 5 pixels ("Tools" + "Radio coverage" + "Combined cartesian"), distinct locations are evaluated for coverage within the map. The CoverageData.txt log file (the text file you get when you perform a radio coverage test) has a distinct column named "Best unit", which determines the site (base station) that provides the best coverage for a given location. Once you determine the best position for the two new sites (question 3), parse the CoverageData.txt and determine:
   a) How many locations have the newly added sites (either of the two new sites) as the "Best unit"?
   b) Calculate the number of locations that are assigned to each new site (i.e., the number of locations that have NewSite1 as "Best unit" and the number of locations that have NewSite2 as "Best unit"). In your opinion, is it better if the new sites you created have approximately the same number of locations or not? Explain.

5) Explain briefly the method you used for deciding where to locate the two new sites (i.e., point out the most important factors in determining the location of new sites).

6) Discuss briefly how the Radio Mobile application could be used to evaluate the carrier-to-interference ratios between two sites operating at the same frequency.